



# BRIGMAN

## Deep Sea Technology

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## Software Requirements Specification (SRS)

### Abyss Navigator Suite Project

### Brigman Deep Sea Technology

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# 1 Introduction

## 1.1 Purpose

This Software Requirements Specification (SRS) defines the software requirements for the **Abyss Navigator Suite**, an unmanned deep-sea submersible fleet control system for Triton Deepwater. It is intended to be the baseline agreement for design, implementation, verification, and change control.

## 1.2 Document Conventions

**Template source.** This SRS is structured using the *Software Requirements Specification* template by **Karl E. Wiegiers (1999)** (provided in *References/srs\_template-ieee.pdf*), adapted to incorporate the course rubric requirements (codification, iRTM, and change mitigation).

**Normative language.** All requirements use **shall** to indicate mandatory behavior.

**Codification (mandatory).** The following ID schemes are used throughout:

- **Software requirements:** <DOMAIN>-<FEATURE>-<NNN> (e.g., SAF-EMG-001).
- **Technology requirements:** TECH-<AREA>-<NNN> (e.g., TECH-PLAT-001).
- **Quality model features:** QUAL-<AREA>-<NNN> (e.g., QUAL-SAF-001).
- **iRTM entries:** RTM-<NNN> (e.g., RTM-001).
- **Change mitigation controls:** CHG-<AREA>-<NNN> (e.g., CHG-CTRL-001).

**Priority levels.** Priority is recorded as **Essential**, **Desirable**, or **Acceptable** to align with the Project Specification.

## 1.3 Intended Audience and Reading Suggestions

This SRS is intended for Triton Deepwater stakeholders, the Brigman development and verification teams, project management, and independent auditors. Readers should begin with Sections 1–3 (scope, constraints, codification) before using Section 4 (requirements) and Sections 7–8 (traceability and change mitigation).

## 1.4 Product Scope

The Abyss Navigator Suite enables unmanned missions executed autonomously, remotely supervised from a surface or mothership environment, with controlled intervention, safe emergency handling, and operational traceability.

## 1.5 References

- Triton Deepwater, *Project Specification: The Abyss Navigator Suite*, ICT2113, 2026.
- Brigman Deep Sea Technology, *Software Requirements Analysis Report (BCIC Methodology)*, 2026.
- Kevin Anthony Jones, *Laborial Guide for Class Project: Reqs now & near future*, Dec 2025.
- Kevin Anthony Jones, *Guide Rubric for Assessment of Deliverables*, Jan 2026.
- Karl E. Wiegiers, *Software Requirements Specification Template*, 1999.

# 2 Overall Description

## 2.1 Product Functions

At a high level, the system provides:

- Mission planning and autonomous execution for unmanned deep-sea missions.
- Remote monitoring, operator alerting, and controlled remote intervention.
- Fleet-level supervision for multiple submersibles concurrently.
- Emergency detection, safe-state transition, and recovery coordination.
- Secure operational data management, logging, and post-mission traceability.

## 2.2 User Classes and Characteristics

- **Mission Supervisor:** monitors mission progress and authorizes intervention.
- **Fleet Operator:** manages multi-vehicle assignments and scheduling.
- **Safety & Compliance Officer:** reviews anomalies, audits logs, and ensures governance obligations are met.
- **System Administrator:** manages credentials, configuration, and system health.
- **Data Analyst:** exports mission data for downstream analysis.

## 2.3 Operating Environment (Latest Tech)

The operating environment is characterized by remote supervision from a mothership environment and constrained connectivity to submersibles. The vendor-selected software platform is described in Section 5. Key operating constraints include:

- Underwater communications may be intermittent and low-bandwidth.
- Extreme pressure and low visibility increase reliance on sensor-driven autonomy.
- Operational data is safety- and audit-relevant and must be protected.

## 2.4 Design and Implementation Constraints

The following constraints apply:

- **Remote operations context:** mission supervision and control are performed remotely from surface/mothership environments.
- **Communications constraints:** the system must operate under constrained bandwidth/latency and intermittent connectivity.
- **Security and governance:** operational data protection and auditability are required.
- **Safety-first policy:** failures and abnormal conditions must bias toward safe-state behavior.

## 2.5 User Documentation

The following documentation shall be delivered with the software:

- Operator manual (mission planning, supervision, interventions, safety procedures).
- System administration guide (credentials, configuration, deployment).
- Interface/API guide for integrating Triton-provided sensors and external systems.

## 2.6 Assumptions and Dependencies

- Triton provides mission sensors and their interface specifications for integration.
- Missions are supervised by trained and authorized personnel.
- Regulatory obligations apply to maritime safety and data governance.

### 3 Codification

#### 3.1 Codification Rules

**Requirement IDs** are stable across revisions. The numeric suffix increments only when a new requirement is added; edited requirements retain their IDs to preserve traceability in the iRTM.

#### 3.2 Requirement Attributes and Quality Tagging

Each requirement in Section 4 is recorded with:

- **Source:** originating Project Specification clause(s) (e.g., BO-L1-4, IT-L0-C3).
- **Priority:** Essential / Desirable / Acceptable.
- **Quality tags:** one or more QUAL-\* IDs from Section 6.
- **Verification:** inspection, analysis, test, or demonstration.
- **Use Case link:** reference to the primary use case(s) in Figure 2 (numbered UC01, UC02, ...) that realize or exercise the requirement.

### 4 Specific Requirements

#### 4.1 Navigation & Mission Autonomy

| Req ID        | Priority  | Source               | Quality Tags                  | Requirement                                                                                                                                                                                          | Verification                                                         | Use Case Link                                       |
|---------------|-----------|----------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------|
| NAV-PLAN-001  | Essential | BO-L1-1              | QUAL-OPS-001,<br>QUAL-TRC-001 | The system shall allow an authorized operator to create and version a mission plan containing an ordered set of waypoints (latitude, longitude, depth) and mission phases.                           | Demo: Create/edit/version plan; inspect stored plan versions.        | <a href="#">UC01</a> (Mission planning)             |
| NAV-EXEC-001  | Essential | BO-L1-1,<br>BO-L0-C1 | QUAL-RES-001,<br>QUAL-SAF-001 | The system shall execute a validated mission plan autonomously without continuous human control for a full mission cycle in a representative simulation.                                             | Test: End-to-end mission simulation with no operator commands.       | <a href="#">UC02</a> (Autonomous mission execution) |
| NAV-WPT-001   | Essential | BO-L1-1              | QUAL-OPS-001                  | The system shall achieve <b>99% waypoint arrival success</b> in which each reached waypoint is within $\leq 5$ m horizontal and $\leq 2$ m vertical error under representative sea-trial conditions. | Analysis: Compare logged track vs. waypoints across trials.          | <a href="#">UC02</a> (Waypoint navigation)          |
| NAV-FLEET-001 | Desirable | BO-L1-3,<br>IT-L1-3  | QUAL-OPS-001,<br>QUAL-RES-001 | The system shall support concurrent supervision of at least <b>3</b> submersibles, including assignment of mission plans and real-time status visibility per vehicle.                                | Test: Simulated 3-vehicle session; verify assignment and visibility. | <a href="#">UC03</a> (Supervise multiple vehicles)  |

#### 4.2 Communications & Remote Operations

| Req ID      | Priority  | Source            | Quality Tags               | Requirement                                                                                                                                                                                                                                                               | Verification                                                                  | Use Case Link                                |
|-------------|-----------|-------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------|
| COM-MON-001 | Essential | BO-L1-2, IT-L1-1  | QUAL-OPS-001               | The system shall provide remote mission and vehicle status telemetry updates at least once every <b>30 seconds</b> when connectivity is available.                                                                                                                        | Test: Measure telemetry update intervals under nominal link.                  | <a href="#">UC04</a> (Monitor telemetry)     |
| COM-INT-001 | Essential | IT-L1-2, BO-L1-2  | QUAL-OPS-001, QUAL-SAF-001 | The system shall allow an authorized operator to issue remote intervention commands (pause, resume, safe-state) and receive acknowledgement within $\leq 60$ seconds when the link latency is $\leq 30$ seconds.                                                          | Test: Command/ack timing under simulated 30s latency.                         | <a href="#">UC05</a> (Remote intervention)   |
| COM-DEG-001 | Essential | IT-L0-C2, IT-L1-6 | QUAL-RES-001, QUAL-SAF-001 | The system shall support safe operation under constrained communications, including operation at $\geq 1$ kbps bandwidth, $\leq 30$ s one-way latency, and intermittent connectivity, and shall enter safe-state if connectivity is lost continuously for $> 10$ minutes. | Test: Fault-inject bandwidth, latency, and outage; verify safe-state trigger. | <a href="#">UC06</a> (Handle degraded comms) |

### 4.3 Safety & Recovery

| Req ID        | Priority  | Source             | Quality Tags               | Requirement                                                                                                                                                                                                       | Verification                                                      | Use Case Link                           |
|---------------|-----------|--------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------|
| SAF-EMG-001   | Essential | BO-L1-4, BO-L2-4.1 | QUAL-SAF-001, QUAL-OPS-001 | The system shall detect pressure deviations $> \pm 5\%$ from nominal setpoint within <b>5 seconds</b> .                                                                                                           | Test: Inject simulated pressure spike; measure detection latency. | <a href="#">UC07</a> (Detect anomalies) |
| SAF-EMG-002   | Essential | BO-L1-4, BO-L2-4.1 | QUAL-SAF-001, QUAL-OPS-001 | The system shall detect temperature deviations $> \pm 10\%$ from nominal within <b>10 seconds</b> .                                                                                                               | Test: Thermal chamber simulation or sensor-in-the-loop test.      | <a href="#">UC07</a> (Detect anomalies) |
| SAF-EMG-003   | Essential | BO-L1-4, BO-L2-4.1 | QUAL-SAF-001, QUAL-OPS-001 | The system shall detect power supply deviations $> \pm 15\%$ from nominal within <b>10 seconds</b> .                                                                                                              | Test: Power supply fault injection; measure detection latency.    | <a href="#">UC07</a> (Detect anomalies) |
| SAF-STATE-001 | Essential | BO-L2-4.2          | QUAL-SAF-001               | Upon detection of any abnormal condition, the system shall transition the submersible to a <b>safe operational state</b> defined as neutral buoyancy, minimal power consumption, and active communication beacon. | Integration test: End-to-end emergency scenario.                  | <a href="#">UC08</a> (Enter safe state) |

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| Req ID       | Priority  | Source  | Quality Tags                  | Requirement                                                                                                                                                                                                         | Verification                                                                  | Use Case Link                                    |
|--------------|-----------|---------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------|
| SYS-CONT-001 | Essential | BO-L1-5 | QUAL-RES-001,<br>QUAL-SAF-001 | When non-critical degradation occurs, the system shall continue the mission in a degraded mode while alerting the operator and logging the degradation event.                                                       | Test: Inject non-critical fault; confirm degraded continuation and alert/log. | <a href="#">UC09</a> (Continue in degraded mode) |
| SYS-TERM-001 | Essential | BO-L1-5 | QUAL-SAF-001,<br>QUAL-TRC-001 | When termination is required, the system shall terminate the mission in an orderly manner by completing the current operation when safe, logging the shutdown reason, and entering safe-state.                      | Test: Inject critical fault; verify ordered termination sequence.             | <a href="#">UC10</a> (Terminate mission)         |
| SYS-REC-001  | Essential | BO-L1-5 | QUAL-SAF-001,<br>QUAL-OPS-001 | After entering safe-state, the system shall support recovery coordination by continuously broadcasting a communication beacon and exposing last-known position and status to operators until recovery is confirmed. | Demo/Test: Simulate safe-state; verify beacon and status visibility.          | <a href="#">UC11</a> (Coordinate recovery)       |

#### 4.4 Data Logging & Accountability

| Req ID       | Priority  | Source                  | Quality Tags                  | Requirement                                                                                                                                                              | Verification                                                         | Use Case Link                                  |
|--------------|-----------|-------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|------------------------------------------------|
| DATA-LOG-001 | Essential | BO-L1-6,<br>IT-L1-5     | QUAL-TRC-001                  | The system shall log mission events and operator actions with timestamp (UTC), vehicle ID, and operator ID attribution.                                                  | Inspection/Test: Review log records for required fields.             | <a href="#">UC12</a> (Record operations)       |
| DATA-LOG-002 | Essential | IT-L1-5,<br>IT-L0-C3    | QUAL-SEC-001,<br>QUAL-TRC-001 | The system shall store operational logs in an append-only form and retain them for at least <b>5 years</b> to support audit and investigation.                           | Inspection: Verify storage configuration and retention policy.       | <a href="#">UC12</a> (Record operations)       |
| DATA-LOG-003 | Essential | BO-L2-6.2,<br>IT-L2-5.1 | QUAL-OPS-001,<br>QUAL-TRC-001 | The system shall allow authorized retrieval and export of mission records for a $\leq 24$ -hour mission into a structured format (CSV and JSON) within $\leq 2$ minutes. | Test: Export timed retrieval under representative data volume.       | <a href="#">UC13</a> (Retrieve/export records) |
| DATA-LOG-004 | Desirable | BO-L1-7,<br>BO-L2-6.2   | QUAL-OPS-001                  | The system shall generate a post-mission summary report (mission timeline, anomalies, interventions) within $\leq 10$ minutes after mission completion.                  | Test: Complete mission; measure report generation time and contents. | <a href="#">UC14</a> (Post-mission summary)    |

#### 4.5 External Interface Requirements



| Req ID       | Priority  | Source   | Quality Tags | Requirement                                                                                                                                                                                                                 | Verification                                                       | Use Case Link                             |
|--------------|-----------|----------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------|
| INT-SENS-001 | Essential | IT-L0-A1 | QUAL-RES-001 | The system shall integrate customer-supplied mission sensors using <b>ROS2 middleware</b> and <b>DDS</b> transport, and shall ingest sensor messages in a documented <b>JSON</b> schema defined in the Interface/API guide. | Test: Sensor-in-the-loop integration against the published schema. | <a href="#">UC15</a> (Sensor integration) |

#### 4.6 Security & Governance

| Req ID       | Priority  | Source             | Quality Tags               | Requirement                                                                                                                                                | Verification                                                                | Use Case Link                                |
|--------------|-----------|--------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------|
| SEC-ENC-001  | Essential | IT-L0-C3, IT-L1-4  | QUAL-SEC-001               | The system shall protect operational data using <b>AES-256</b> encryption for data at rest and for encrypted payloads in transit.                          | Inspection/Test: Verify crypto configuration and encrypted traffic capture. | <a href="#">UC16</a> (Protect data)          |
| SEC-AUTH-001 | Essential | IT-L0-C3, IT-L1-4  | QUAL-SEC-001               | The system shall enforce mutual authentication for system-to-system communications using <b>X.509 certificates</b> .                                       | Test: Reject invalid/expired certificates; accept valid certificates only.  | <a href="#">UC16</a> (Mutual authentication) |
| SEC-AUD-001  | Essential | IT-L0-C3, BO-L0-C4 | QUAL-SEC-001, QUAL-TRC-001 | The system shall maintain audit logs for access to operational data and for administrative configuration changes, including actor identity and timestamps. | Inspection/Test: Perform actions; confirm corresponding audit records.      | <a href="#">UC17</a> (Audit access)          |

### 5 Technology Requirements (Latest Tech)

The following technology requirements define the intended operating platform, data store, and interface patterns selected by Brigman Deep Sea Technology to satisfy the Project Specification constraints.

| Tech ID       | Area       | Specification                                                                                                                                                                               |
|---------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TECH-PLAT-001 | Platform   | Vehicle-side software components shall run on a Linux-based OS with ROS2 middleware support; mothership-side services shall support deployment on standard workstation/server environments. |
| TECH-DB-001   | Database   | The system shall store operational and mission records in a structured database that supports retention policies and encrypted storage (e.g., PostgreSQL or equivalent).                    |
| TECH-NET-001  | Networking | The system shall support message transport compatible with constrained links (DDS where available) and shall support store-and-forward buffering during outages.                            |
| TECH-API-001  | APIs       | External and internal service interfaces shall use versioned JSON schemas and provide a documented API contract for integration and export.                                                 |

Table 7: Technology requirements for operating environment and platform

## 6 Quality Development Model

Brigman Deep Sea Technology applies the **BDST Quality Model v1.0** to ensure requirements are not only functional but also safe, secure, and verifiable under operational constraints. Each requirement in Section 4 is tagged with one or more quality feature IDs below.

| Quality ID   | Feature Name                  | What it Covers / Measures                                                                                                                                                                                           |
|--------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QUAL-SAF-001 | Safety Integrity              | Measures correct detection of abnormal conditions, safe-state transitions, and recovery behavior under fault injection.                                                                                             |
| QUAL-SEC-001 | Security & Governance         | Measures encryption coverage, authentication strength, audit completeness, and unauthorized access resistance.                                                                                                      |
| QUAL-RES-001 | Resilience under Constraints  | Measures mission continuity and safe behavior during comm outages, bandwidth limits, and sensor degradation.                                                                                                        |
| QUAL-TRC-001 | Traceability & Accountability | Measures ability to reconstruct mission timelines and decisions from logs with actor attribution and retention.                                                                                                     |
| QUAL-OPS-001 | Operability & Performance     | Measures operator ability to plan missions, interpret telemetry/alerts, intervene reliably, and meet timing/performance thresholds (telemetry update rates, detection latency, report generation, retrieval times). |

Table 8: BDST Quality Model v1.0 (quality feature names and measurement intent)

## 7 Integrated Requirements Traceability Matrix (iRTM)

The iRTM traces each Project Specification clause to the corresponding SRS requirement(s) or SRS section(s), using minimized paraphrases to preserve CReAM meaning while remaining concise.

Table 9: iRTM: Project Specification (PS) to SRS traceability

| iRTM ID | PS Ref   | PS Clause (min.)                              | SRS Element                                   | SRS Clause / Coverage (min.)                                     |
|---------|----------|-----------------------------------------------|-----------------------------------------------|------------------------------------------------------------------|
| RTM-001 | BO-L0-C1 | Unmanned operations at extreme depths.        | NAV-EXEC-001                                  | Autonomous execution without continuous human control.           |
| RTM-002 | BO-L0-C2 | Safety-first: eliminate human exposure.       | SAF-STATE-001;<br>SYS-TERM-001                | Safe-state bias and orderly termination under risk.              |
| RTM-003 | BO-L0-C3 | Extreme pressure/low visibility/limited comm. | COM-DEG-001;<br>SAF-EMG-001                   | Constrained-link behavior; abnormal detection.                   |
| RTM-004 | BO-L0-C4 | Comply with maritime safety/data governance.  | SEC-AUD-001;<br>DATA-LOG-002                  | Auditable access and retained records.                           |
| RTM-005 | BO-L0-A1 | Missions supervised by trained personnel.     | Section 2.7                                   | Operational assumption recorded (software supports supervision). |
| RTM-006 | BO-L1-1  | Autonomous mission execution required.        | NAV-PLAN-001;<br>NAV-EXEC-001;<br>NAV-WPT-001 | Plan, autonomous execution, and waypoint performance.            |

Continued on next page

| iRTM ID | PS Ref    | PS Clause (min.)                                 | SRS Element                                   | SRS Clause / Coverage (min.)                               |
|---------|-----------|--------------------------------------------------|-----------------------------------------------|------------------------------------------------------------|
| RTM-007 | BO-L1-2   | Remote mission oversight required.               | COM-MON-001;<br>COM-INT-001                   | Telemetry updates and remote interventions.                |
| RTM-008 | BO-L1-3   | Coordinated multi-submersible operation.         | NAV-FLEET-001                                 | Concurrent supervision and mission assignment.             |
| RTM-009 | BO-L1-4   | Handle abnormal/emergency conditions.            | SAF-EMG-001..003;<br>SAF-STATE-001            | Detection thresholds and safe-state transition.            |
| RTM-010 | BO-L1-5   | Continue safely or terminate orderly.            | SYS-CONT-001;<br>SYS-TERM-001;<br>SYS-REC-001 | Degraded continuation, termination, recovery coordination. |
| RTM-011 | BO-L1-6   | Accountability for mission activities/decisions. | DATA-LOG-001;<br>SEC-AUD-001                  | Operator-attributed logging and auditability.              |
| RTM-012 | BO-L1-7   | Outputs usable for ops and analytics.            | DATA-LOG-003;<br>DATA-LOG-004                 | Exportable records and summary reporting.                  |
| RTM-013 | BO-L2-4.1 | Abnormal conditions identifiable.                | SAF-EMG-001..003                              | Defined abnormal thresholds and detection timing.          |
| RTM-014 | BO-L2-4.2 | Transition to safe state on critical.            | SAF-STATE-001                                 | Defined safe operational state behavior.                   |
| RTM-015 | BO-L2-6.1 | Activities/decisions traceable.                  | DATA-LOG-001;<br>SEC-AUD-001                  | Trace logs and audit logs with actor identities.           |
| RTM-016 | BO-L2-6.2 | Records support review/investigation.            | DATA-LOG-002..004                             | Retention, export, and post-mission reporting.             |
| RTM-017 | IT-L0-C1  | Remote supervision/control context.              | COM-MON-001;<br>COM-INT-001                   | Remote monitoring and intervention functions.              |
| RTM-018 | IT-L0-C2  | Constrained bandwidth/latency/intermittent.      | COM-DEG-001                                   | Explicit bandwidth, latency, and outage thresholds.        |
| RTM-019 | IT-L0-C3  | Protect operational data per obligations.        | SEC-ENC-001;<br>SEC-AUTH-001;<br>SEC-AUD-001  | Encryption, authentication, audit.                         |
| RTM-020 | IT-L0-A1  | Customer provides sensors for integration.       | INT-SENS-001                                  | ROS2/DDS/JSON-based sensor integration.                    |
| RTM-021 | IT-L1-1   | Remote monitoring capability required.           | COM-MON-001                                   | Telemetry update rate requirement.                         |
| RTM-022 | IT-L1-2   | Remote intervention capability required.         | COM-INT-001                                   | Remote command/ack requirement.                            |
| RTM-023 | IT-L1-3   | Multi-vehicle supervision required.              | NAV-FLEET-001                                 | 3-vehicle concurrent supervision.                          |
| RTM-024 | IT-L1-4   | Protect confidentiality/integrity of data.       | SEC-ENC-001;<br>SEC-AUTH-001                  | Encryption and mutual authentication.                      |
| RTM-025 | IT-L1-5   | Logging mission events and operator actions.     | DATA-LOG-001..004                             | Logging, retention, export, reporting.                     |

Continued on next page

| iRTM ID | PS Ref    | PS Clause (min.)                           | SRS Element                   | SRS Clause / Coverage (min.)                   |
|---------|-----------|--------------------------------------------|-------------------------------|------------------------------------------------|
| RTM-026 | IT-L1-6   | Safe operation under degraded comm.        | COM-DEG-001;<br>SYS-CONT-001  | Degraded comm handling and mission continuity. |
| RTM-027 | IT-L2-5.1 | Logs sufficient for review/accountability. | DATA-LOG-003;<br>DATA-LOG-004 | Exportable records and review-ready reports.   |

## 8 Change Mitigation Protocol

This protocol reduces downstream SDLC errors by preventing uncontrolled requirement drift, ensuring every change is evaluated for impact on implementation, verification, and traceability.

| Change ID    | Control / Strategy                                                                                                                                                           |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CHG-CTRL-001 | <b>Baseline control:</b> the SRS is baselined at Version 1.0; all changes require a change request (CR) recorded in the repository.                                          |
| CHG-CTRL-002 | <b>Change classification:</b> changes are classified as Editorial, Minor (no external impact), or Major (scope/safety/security/performance impact).                          |
| CHG-CTRL-003 | <b>Impact analysis:</b> each CR must list affected requirement IDs, affected iRTM entries, verification updates, and schedule/risk impacts before approval.                  |
| CHG-CTRL-004 | <b>Approval gates:</b> Major changes require approval by the Customer-designated approver and the Vendor authenticator; Minor changes require Vendor authenticator approval. |
| CHG-CTRL-005 | <b>Traceability maintenance:</b> any approved change must update Section 4 and Section 7 in the same revision to keep the iRTM consistent.                                   |
| CHG-CTRL-006 | <b>Verification alignment:</b> changes that affect timing, safety, or security requirements must include updated test cases and acceptance criteria prior to implementation. |

Table 10: Change mitigation controls (codified)

## 9 Addenda

| Addendum ID | Statement                                                                                                                                                                             |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADD-001     | <b>Project direction baseline:</b> This SRS assumes the customer direction change to a fully unmanned, autonomous-first operating model is in force for all missions.                 |
| ADD-002     | <b>Cost redaction note:</b> Any cost figures referenced from the Project Specification are treated as indicative and may be redacted in released versions to avoid biasing proposals. |

Table 11: Addenda (purpose-appropriate notes adopted from the Project Specification)

## 10 Requirements Modeling

The following models are included to support stakeholder understanding and to maintain traceability between requirements and operational context. Refer to Figure 2 (Use Case Diagram) for the consolidated actor–system interactions used throughout this SRS.

Additional Project Specification models (e.g., organization chart and roles/responsibilities) are adopted by reference from the Project Specification appendices to avoid divergence across artifacts.

## 11 Sign-off

### 11.1 Vendor Authentication

| Name                                           | Employee ID    | Signature                                                                           |
|------------------------------------------------|----------------|-------------------------------------------------------------------------------------|
| Muhammad Ridwan Putra<br>Jasni (Authenticator) | BDST-RE-BA-001 |  |

Table 12: Vendor authenticator sign-off (signature provided)

### 11.2 Designated Approvers (No Signatures Required)

| Organization                | Role                                   | Name (Designated) |
|-----------------------------|----------------------------------------|-------------------|
| Customer (Triton Deepwater) | Project Leader / Requirements Approver | Sarah Whitman     |
| Customer (Triton Deepwater) | Safety & Compliance Approver           | (TBD)             |
| Vendor (Brigman)            | Engineering Delivery Lead              | (TBD)             |
| Vendor (Brigman)            | Verification & Validation Lead         | (TBD)             |

Table 13: Designated approvers (placeholders where not yet nominated)

## Repository Information

| PROJECT REPOSITORY                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="https://github.com/Spooky312/The-Abyss-Navigation-Suite-INF2113">https://github.com/Spooky312/The-Abyss-Navigation-Suite-INF2113</a><br><i>SRS, models, and traceability artifacts maintained under version control</i> |

## A Glossary

- **BCIC:** Breakdown, Clarify, Interpret, Categorize.
- **CRaM:** Correct, Relevant, and Measurable requirements organization principle.
- **iRTM:** Integrated Requirements Traceability Matrix.
- **Safe-state:** neutral buoyancy, minimal power, beacon active.

## B To Be Determined (TBD) List

- Customer nomination of Safety & Compliance approver name.
- Final selection of database product/version for TECH-DB-001 (if different from PostgreSQL-equivalent).
- Final JSON schemas for INT-SENS-001 (to be published with the Interface/API guide).

## C Figures

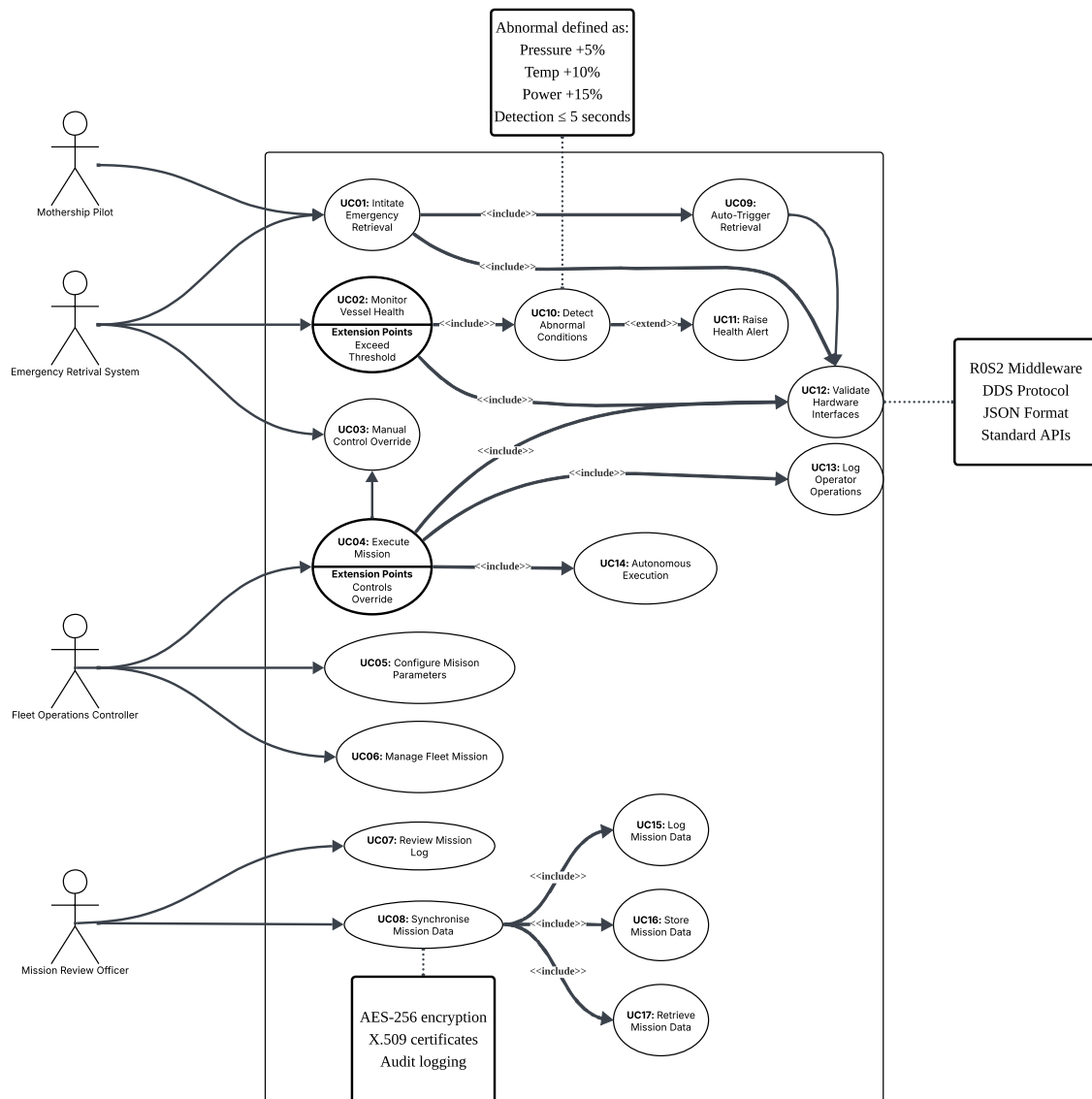


Figure 1: Use Case Diagram

## Instructions

The diagram illustrates the primary actors (surface controller, mothership crew, autonomous submersible) and their interactions with major system functions such as mission planning, launch/recovery, sensor telemetry, fault handling, and data logging. Follow each actor's swim lane to see how the mission progresses from preparation to post-mission data review; the colored connectors highlight safety-critical flows (emergency surfacing, manual override) that must remain available throughout operations.

## Appendix A: Response Sequence

| Use Case No.                                    | Use Case Flow (Description)                                                                                                                                                                                                                                                                                                                                                   | Use Case End Goal                                               |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>UC01 – Initiate Emergency Retrieval</b>      | <b>Main Flow:</b><br><ol style="list-style-type: none"><li>1. Pilot detects an emergency situation.</li><li>2. Pilot selects the emergency retrieval command.</li><li>3. System validates hardware interfaces.</li><li>4. System activates retrieval protocol.</li><li>5. Vessel begins return-to-surface procedure.</li></ol>                                                | The vessel is safely retrieved during an emergency.             |
| <b>UC02 – Monitor Vessel Health</b>             | <b>Main Flow:</b><br><ol style="list-style-type: none"><li>1. System continuously receives vessel telemetry.</li><li>2. System checks pressure, temperature, and power levels.</li><li>3. System compares readings with safety thresholds.</li><li>4. System detects abnormal readings if thresholds are exceeded.</li><li>5. System prepares alerts for operators.</li></ol> | Vessel health is monitored continuously to detect issues early. |
| <b>UC03 – Manual Control Override</b>           | <b>Main Flow:</b><br><ol style="list-style-type: none"><li>1. Operator selects the manual override option.</li><li>2. System verifies operator authorization.</li><li>3. System disables autonomous control temporarily.</li><li>4. Operator sends manual navigation commands.</li></ol>                                                                                      | Operator gains direct control of the vessel when necessary.     |
| <b>UC04 – Execute Mission Controls Override</b> | <b>Main Flow:</b><br><ol style="list-style-type: none"><li>1. System receives manual control commands.</li><li>2. System validates command parameters.</li><li>3. System applies override to mission controls.</li><li>4. System logs operator actions.</li><li>5. Vessel executes the override instructions.</li></ol>                                                       | Mission continues under manual control safely.                  |
| <b>UC05 – Configure Mission Parameters</b>      | <b>Main Flow:</b><br><ol style="list-style-type: none"><li>1. Fleet Controller opens mission configuration interface.</li><li>2. Controller enters mission parameters (depth, path, duration).</li><li>3. System validates parameter limits.</li><li>4. Controller confirms configuration.</li><li>5. System stores mission parameters.</li></ol>                             | Mission parameters are prepared for safe mission execution.     |
| <b>UC06 – Manage Fleet Mission</b>              | <b>Main Flow:</b><br><ol style="list-style-type: none"><li>1. Fleet Controller views fleet dashboard.</li><li>2. Controller assigns missions to vessels.</li><li>3. System distributes mission data to vessels.</li><li>4. Controller monitors fleet progress.</li><li>5. System updates fleet status in real time.</li></ol>                                                 | Multiple vessels are coordinated efficiently.                   |
| <b>UC07 – Review Mission Log</b>                | <b>Main Flow:</b><br><ol style="list-style-type: none"><li>1. Mission Review Officer accesses mission log system.</li><li>2. Officer selects a completed mission.</li><li>3. System retrieves stored mission records.</li><li>4. System displays full mission timeline.</li></ol>                                                                                             | Mission performance is evaluated using stored logs.             |

Continued on next page



| Use Case No.                               | Use Case Flow (Description)                                                                                                                                                                                                                                                                                                                               | Use Case End Goal                                           |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>UC08 – Synchronise Mission Data</b>     | <b>Main Flow:</b> <ol style="list-style-type: none"> <li>1. System collects mission data from vessels.</li> <li>2. System encrypts data using secure protocols.</li> <li>3. System synchronizes data with control station servers.</li> <li>4. System verifies successful synchronization.</li> <li>5. System updates mission data repository.</li> </ol> | Mission data remains consistent across all systems.         |
| <b>UC09 – Auto-Trigger Retrieval</b>       | <b>Main Flow:</b> <ol style="list-style-type: none"> <li>1. System detects abnormal operating conditions.</li> <li>2. System evaluates emergency thresholds.</li> <li>3. System automatically triggers retrieval protocol.</li> <li>4. Vessel receives retrieval command.</li> <li>5. Vessel begins safe recovery procedure.</li> </ol>                   | Automatic safety retrieval occurs during critical failures. |
| <b>UC10 – Detect Abnormal Conditions</b>   | <b>Main Flow:</b> <ol style="list-style-type: none"> <li>1. System continuously monitors sensor data.</li> <li>2. System evaluates pressure, temperature, and power metrics.</li> <li>3. System compares metrics to defined limits.</li> <li>4. System identifies abnormal condition.</li> <li>5. System notifies alert subsystem.</li> </ol>             | Abnormal system conditions are detected quickly.            |
| <b>UC11 – Raise Health Alert</b>           | <b>Main Flow:</b> <ol style="list-style-type: none"> <li>1. System receives abnormal condition signal.</li> <li>2. System generates health alert notification.</li> <li>3. Alert is displayed on operator dashboard.</li> <li>4. System logs the alert event.</li> <li>5. Operator reviews and decides next action.</li> </ol>                            | Operators are notified of system health issues.             |
| <b>UC12 – Validate Hardware Interfaces</b> | <b>Main Flow:</b> <ol style="list-style-type: none"> <li>1. System checks connected hardware modules.</li> <li>2. System tests communication interfaces.</li> <li>3. System verifies sensor availability.</li> <li>4. System confirms interface compatibility.</li> <li>5. System reports validation status.</li> </ol>                                   | Hardware readiness is verified before operations.           |
| <b>UC13 – Log Operator Operations</b>      | <b>Main Flow:</b> <ol style="list-style-type: none"> <li>1. Operator performs system action.</li> <li>2. System captures action details.</li> <li>3. System records timestamp and user identity.</li> <li>4. System stores log entry securely.</li> <li>5. Audit log is updated.</li> </ol>                                                               | Operator activities are recorded for auditing.              |
| <b>UC14 – Autonomous Execution</b>         | <b>Main Flow:</b> <ol style="list-style-type: none"> <li>1. System loads configured mission parameters.</li> <li>2. Vessel begins autonomous navigation.</li> <li>3. System monitors execution progress.</li> <li>4. System records telemetry data.</li> <li>5. Mission continues until completion or interruption.</li> </ol>                            | Vessel completes mission autonomously.                      |

Continued on next page



| Use Case No.                        | Use Case Flow (Description)                                                                                                                                                                                                                                                                                  | Use Case End Goal                                       |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| <b>UC15 – Log Mission Data</b>      | <b>Main Flow:</b><br><ol style="list-style-type: none"><li>1. System collects mission telemetry data.</li><li>2. System records sensor readings.</li><li>3. System timestamps each data entry.</li><li>4. Data is formatted for storage.</li><li>5. Log records are prepared for database storage.</li></ol> | Mission telemetry data is recorded during operations.   |
| <b>UC16 – Store Mission Data</b>    | <b>Main Flow:</b><br><ol style="list-style-type: none"><li>1. System receives mission log records.</li><li>2. System encrypts stored data.</li><li>3. System writes records to database storage.</li><li>4. System confirms successful storage.</li><li>5. Data repository is updated.</li></ol>             | Mission data is securely stored in the system database. |
| <b>UC17 – Retrieve Mission Data</b> | <b>Main Flow:</b><br><ol style="list-style-type: none"><li>1. Officer requests mission data retrieval.</li><li>2. System searches mission database.</li><li>3. System retrieves requested data.</li><li>4. System formats data for display.</li><li>5. Data is presented to the officer.</li></ol>           | Stored mission data becomes accessible for analysis.    |

## Appendix B: Wireframe